

Advanced Databases

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1 Introduction

1.1 Purpose

The goal of this document is to indicate the approaches made to design a social network with the aim of connecting students and employees based on their interests and skills, specifically the ODM design. This is because due to the nature of the data being handled, a document oriented NoSQL database like MongoDB is a good fit for this use case.

1.2 Toolset

Toolset used for the creation of the ODM:

- MongoDB: as the NoSQL database management system.
- MongoDB Atlas: as the cloud service to host the database.
- Python: as the programming language.
- YAML: for the MongoDB schema definition.
- Nix: for the management of dependencies and environments.
- L^AT_EX: for the writing of documentation.

Even though MongoDB Atlas is used to host the database, the option exists to use a local instance of MongoDB.

2 Conceptual Design

2.1 Schema Design

Due to the database being designed to connect users based on their interests, the main entity is the `User`. Users aren't required to have all the attributes but they are required to have the `email` and `name`, `email` being the unique index.

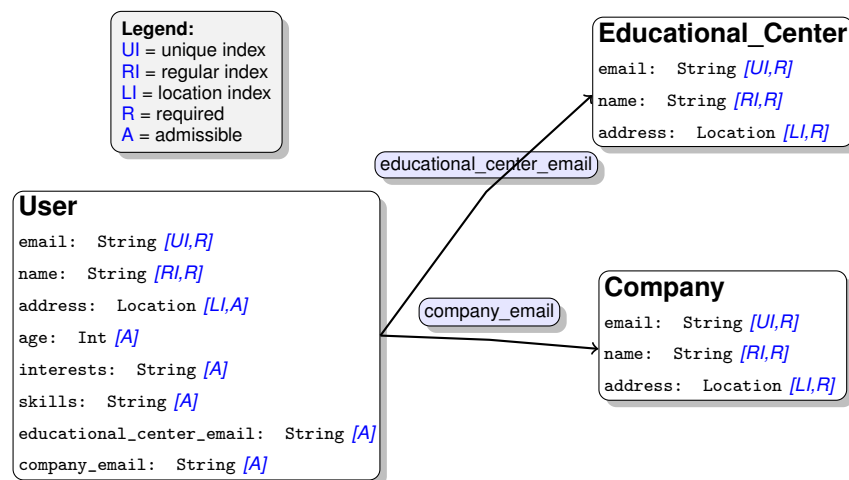
As for the `Company` and `Educational_Center` entities, the first approach considered was to have them as a single collection with a `type` field to specify if it was a company or an educational center. However, it was decided to create two separate collections for them as in the future they might have different

attributes such as degrees offered in the case of educational centers or jobs available in the case of companies. These entities are also required to provide their email, name and also address, since companies and educational centers are required by law to provide their address.

2.2 Schema Overview

Below a diagram showing the schema overview with the entities, their attributes and how they relate to each other is shown. Things to take note of:

- Even though interests and skills are shown as being of type string, in the actual implementation they are strings that contain words separated by commas.
- The educational_center_email and company_email fields in the User entity are used to relate a user to an educational center or a company respectively.



3 ODM Implementation

3.1 Schema Definition

The schemas for the collections were defined using YAML files. These YAML files are then used by Python scripts to create the collections in the database and the appropriate classes in Python to handle them.

The following file is an example file used for testing the model for User.

```

1 User:
2   required_vars:
3     - name
4     - email
5   admissible_vars:
6     - age
7     - address
8   unique_indexes:
9     - name
10  regular_indexes:
11    - email
12  location_index: address

```

Listing 1: models_test.yaml